

GE261004 DESIGN THINKING

Unit III – Ideation and Creative Thinking · Presentation 4 of 6

The SCAMPER Technique

Where SCAMPER came from · the seven prompts · a worked example · running a session · when it works and when it does not

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In this presentation

- Where SCAMPER came from
- The seven prompts
- What each prompt asks
- A worked example
- Running a SCAMPER session
- When it works, when it does not
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- In class: SCAMPER your own idea



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The technique

Seven fixed prompts, applied whether or not you feel inspired

Where SCAMPER came from

- Alex Osborn published a checklist of idea-spurring questions in *Applied Imagination* in 1953
- Bob Eberle turned that checklist into the SCAMPER mnemonic in 1971
- It is a forced-association tool: each letter makes you look at the thing one fixed way
- It works best on something that already exists and needs improving

KEY POINT SCAMPER does not wait for inspiration. It hands you seven prompts and makes you answer all of them.

The seven prompts

S

Substitute

what else instead?

C

Combine

what could be joined?

A

Adapt

what is like this?

M

Modify

bigger, smaller, other?

P

Put to other use

who else could use it?

E

Eliminate

what can be removed?

R

Reverse

what if it ran backwards?

Answer every letter, even the ones that seem to lead nowhere.

What each prompt asks

Letter	What it asks	A second way to ask it
Substitute	What else instead?	Which part could be something else?
Combine	What could be joined?	What if two of these became one?
Adapt	What is like this?	Who else solved a problem like it?
Modify	Bigger, smaller, other?	What if a feature were exaggerated?
Put to use	Who else could use it?	Where else would this work?
Eliminate	What can be removed?	What if the main part were gone?
Reverse	What if it ran backwards?	What if the order were inverted?

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Using it

A worked example, and the sessions where it belongs

Worked example: the campus water cooler

Substitute

A clay pot instead of a plastic tank – cooler water, no electricity, and it is repairable.

Combine

Put the cooler and a bottle-refill counter together, so one queue serves both.

Eliminate

Remove the cup entirely. Everyone carries a bottle, and the waste disappears.

Reverse

Stop making people walk to the water. Bring the water to the corridor instead.

Running a SCAMPER session

- Name the object or process precisely – “the queue at the water cooler”, not “water”
- Take one letter at a time and give it a fixed three minutes
- Write every answer, including the ones that are obviously impractical
- Only after all seven letters are done may anybody evaluate anything

ACTIVITY Twenty-one minutes: three minutes a letter, one letter at a time, with the object named on the board.

When it works, when it does not

Works well for

- An existing product or process
- Teams who have run dry
- Incremental improvement
- Solo work as much as group

Works badly for

- A problem not yet framed
- Blank-page invention
- Anything with no object at its centre
- Rooms that will not write bad ideas

KEY POINT SCAMPER improves something that exists. It will not tell you what the problem is – Unit II does that.

Getting more out of it

- Run it on the user's workaround, not only on the product
- Apply a letter to a different part each time – the object, the queue, the cost
- Combine with a constraint: substitute, but it has to cost a tenth as much
- Keep the impractical answers; they are often one step from a practical one

KEY POINT The impractical answer is usually the one nobody else in the room would have written down.

In class: SCAMPER your own idea

- Take the strongest idea from your team's brainwriting session
- Run all seven letters against it, three minutes each, writing everything
- Mark the three answers that changed the idea most
- Rewrite the idea in one sentence, incorporating those three

ACTIVITY Twenty-five minutes. Each team presents the before and after sentence in thirty seconds.

THINK ABOUT IT

Which SCAMPER letter did your team find hardest – and what does that say?

- Which letter produced nothing at all?
- Was that the object's fault or the team's?
- Which letter produced the answer you liked least?
- Which one would you run first next time?

Summary

- Osborn's 1953 checklist became Eberle's SCAMPER mnemonic in 1971
- Seven prompts: substitute, combine, adapt, modify, put to use, eliminate, reverse
- A forced-association tool – you answer every letter, inspired or not
- Name the object precisely, then give each letter fixed time
- Best on something that exists; useless on an unframed problem
- Keep the impractical answers – they are often one step from useful

Next: Lateral thinking and the Six Thinking Hats

- Vertical and lateral thinking compared
- Four lateral tools, including provocation
- De Bono's six hats and what each one does
- Why parallel thinking beats argument